

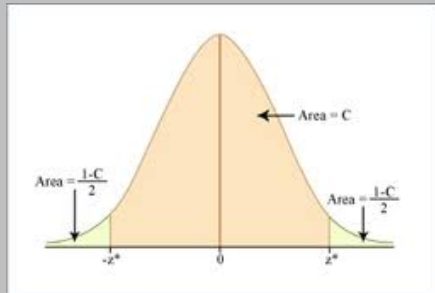
Philosophy of the Social Sciences

TaMoS Lecture #8

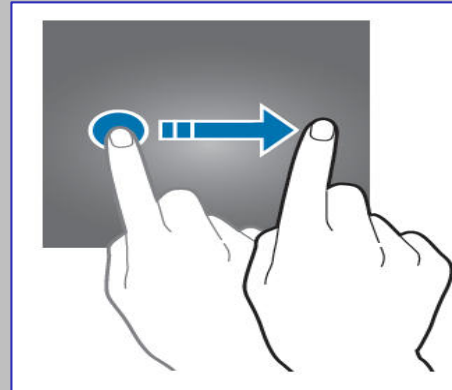
Till Grüne-Yanoff



The Many Social Science Methods



Record choices



How good is
your new
product?



Think-aloud protocol



Simulate behaviour



Observe use
process



Ask evaluation

The Many Social Science Methods

Variety

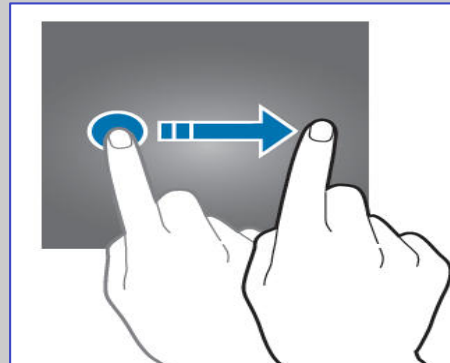
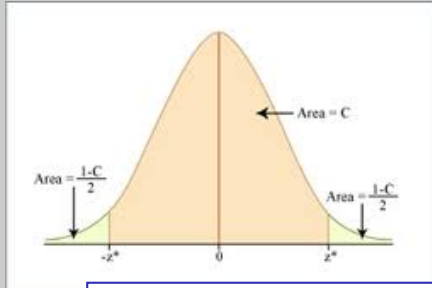
Scientific?

Qualitative

Sampling

Interpretation

Holism



Think cloud protocol

R

How to assess these methods? How to justify choosing between them?



Simulate behaviour

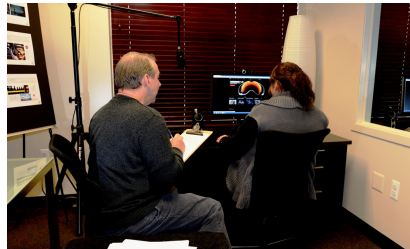


Ask evaluation



Observe use process

What is Special about Social Sciences Methods?



1. Focus on *human behavior* and its origins
2. Investigated by humans through interaction with subjects
observation of behavior – analysis of artefacts – reading/listening to expression –
questioning/conversation – active participation in behavior
3. Production of *qualitative* data
Sound recordings - video recordings - diary entries -experience reports –
photographs – questionnaire answers – open interview answers – documents
4. Analysis of data through *interpretation*

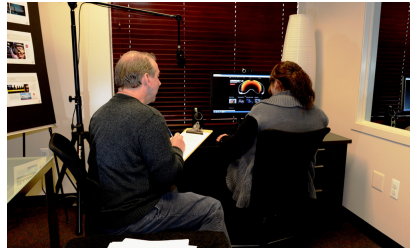
Qualitative Evidence

In what ways does qualitative and quantitative data differ?

- A. Qualitative data contains information about a wide variety of aspects, while quantitative data typically contains information only about a few properties.
- B. Qualitative and quantitative data share no information in common – they have fundamentally different content.
- C. Quantitative approaches require operationalisations; qualitative methods don't
- D. We often use different methods to produce qualitative and quantitative data, respectively.
- E. Quantitative data can be valid, if its measuring methods are reliable. Qualitative data cannot be valid or invalid, but should be evaluated as “credible” or “authentic” instead.

Qualitative and Quantitative Data

D. Production methods often different:



Qualitative and Quantitative Data

B. Data types *not* fundamentally different:
Social science data often can be quantified



Qualitative and Quantitative Data

Social science data often can be quantified

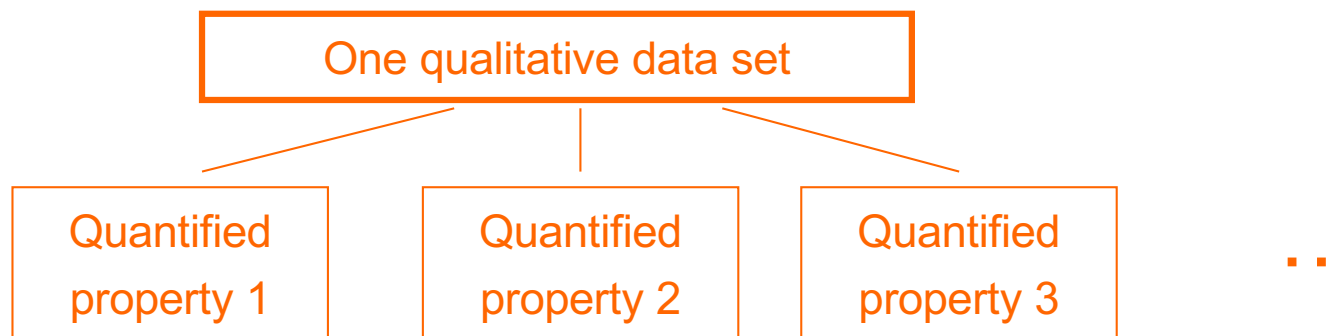
- Prices, demographics, votes,...
- Preferences, beliefs, response scales, networks,...
- Interview coding, statistical music analysis,...

Qualitative and Quantitative Data

A. Quantification often narrows focus



dreamstime.com



Qualitative and Quantitative Data

C. Operationalisation?

Problem of Construct Validity

- e.g. “feeling comfortable”, “being nervous”, “enjoyed the interaction”,...

If construct validity is problematic, there must be some operationalization hidden somewhere...

- Remember: to *operationalize* a feature of interest is to provide a way of linking it to observable properties (lecture 3)
- That also holds for qualitative observations

Qualitative and Quantitative Data

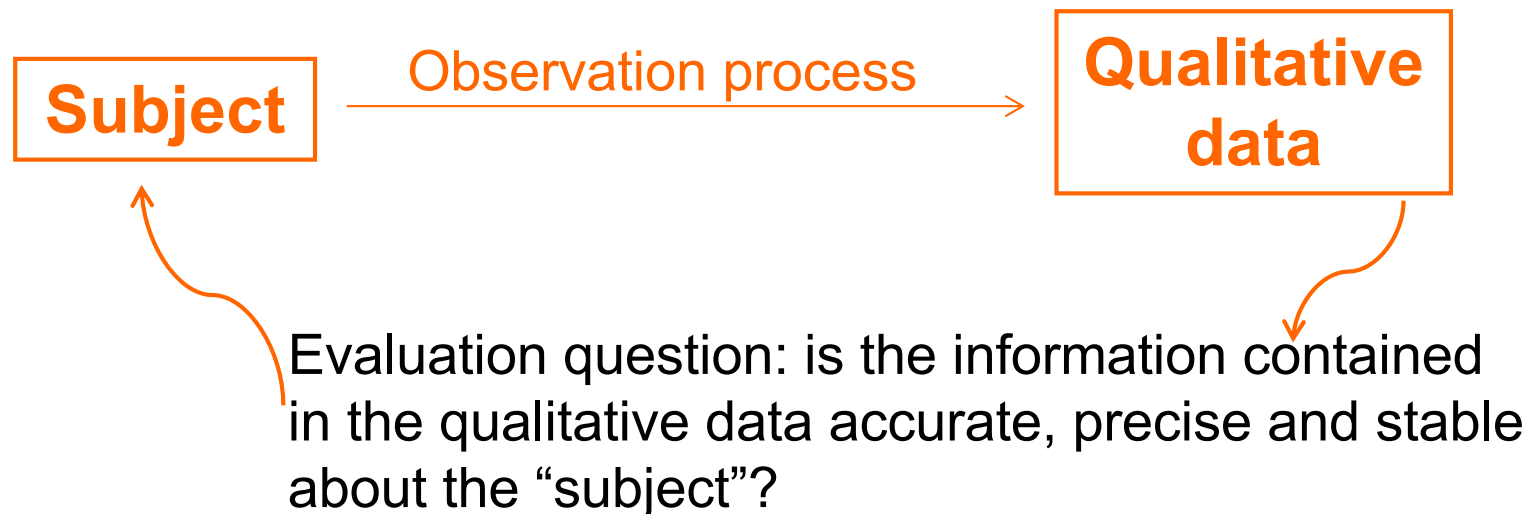
Conditions for good operationalization:

- i. The concept to be observed is defined
- ii. The property of interest is connected with an observable property
- iii. The connection between property of interest and observable property is valid according to our best knowledge
- iv. The connection between property of interest and observable property is sufficiently stable for practical purposes
- v. The observable property is observable with sufficient precision in practice

- For producing quantitative data, one cannot help but make operationalisation explicit
- For producing qualitative data, operationalisation too often remains implicit!

Qualitative and Quantitative Data

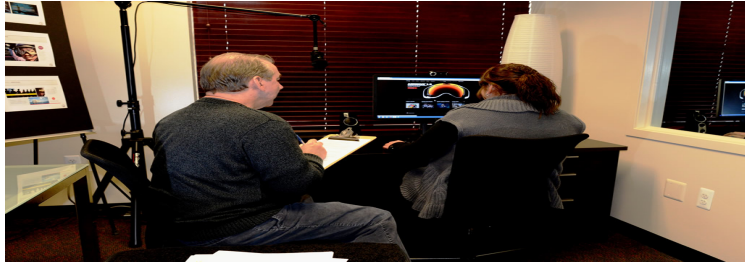
E. Qualitative data can be evaluated as valid or not valid



- Ensure repeatability
- Control for disturbing factors
 - Observer effect
 - Interpretation problem/observer bias
- Show that measure is robust

Interpretation as a Method

What is Interpretation?



What the observer sees:

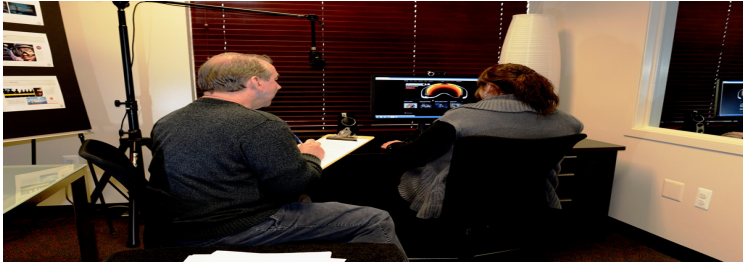


Instruction



Choice

What is Interpretation?



What the subject does:



Instruction

*What does
this mean?*

Understanding

*What are my
choices?
What effects
do my choices
have?*

*Belief
s*

*Do I want
to choose
this way?*

Desires



Choice

What is Interpretation?

- Not just correlating inputs and (behavioral) outputs
- Assume that inputs are mentally processed, leading to behaviour meaningful in the light of subjective understandings, beliefs and desires
- Social scientific interpretation involves the *assignment* of such understandings, beliefs and desires to render observed behaviour meaningful

What is Interpretation?

Variety

Scientific?

Qualitative

Sampling

Interpretation

Holism

Belief-Desire Explanation

Individual i desired state X more than state Y

Individual i believed that action A brought about X but not Y ,
and B Y but not X

Therefore, individual i chose action A over action B

ConceptTest 2

Variety

Scientific?

Qualitative

Sampling

Interpretation

Holism

What makes the following claims true? What reasons do you have for distinguishing these two cases? (max 2 sentences)



A. "The accused murdered the victim by running him over with his car."



B. "Deep Blue wanted to lure Kasparov's queen by offering a pawn."

Time allowed:
2'00

Improving Interpretation

- Minimize interpretation
- Competent observer
- Generate more evidence for uncovering processes:
 - eye tracking
 - thinking aloud protocol
- Check for coherence
 - Repeat questions with slight reformulation
 - *Respondent validation*: subject is allowed to check and amend her answers
- Use intersubjective checks
 - *Inter-rater reliability*: multiple researchers are involved in the analysis
 - *Jury analogy*

The more systematically we employ these strategies, the more confidence we have in the validity of the interpretation!

Naturalism in the Social Sciences

The Naturalist View of Social Science

Social sciences use different methods than the natural science – but they share the same methodology!

- Same goals: explanation & prediction
- Reliability of methods
- Validity of results

The Anti-Naturalist View of Social Science

"If it's social it can't be scientific!"

The social sciences cannot and should not follow the methodology of the natural sciences.

Fundamental differences in methods:

- Qualitative v. quantitative
- Interpretation v. causal regularities

But...as I showed, these differences are more apparent than real!

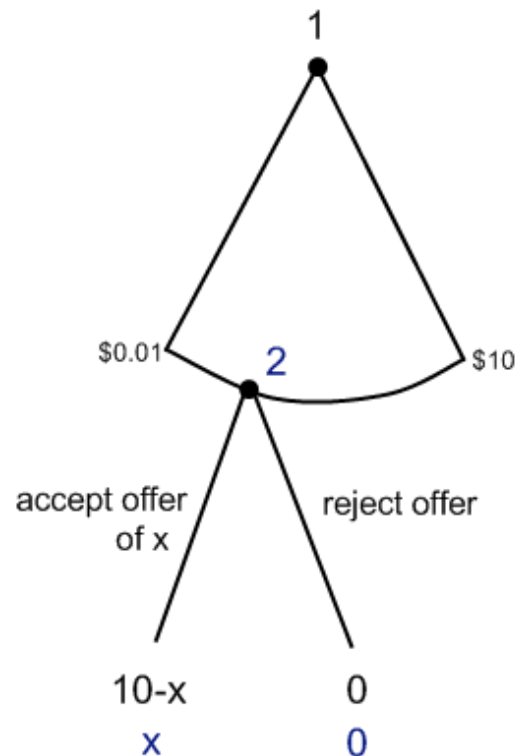
Holism

Methodological Individualism

All good social theory is reducible to individualist accounts.

When Individualists Accounts are Incomplete

The Ultimatum Game



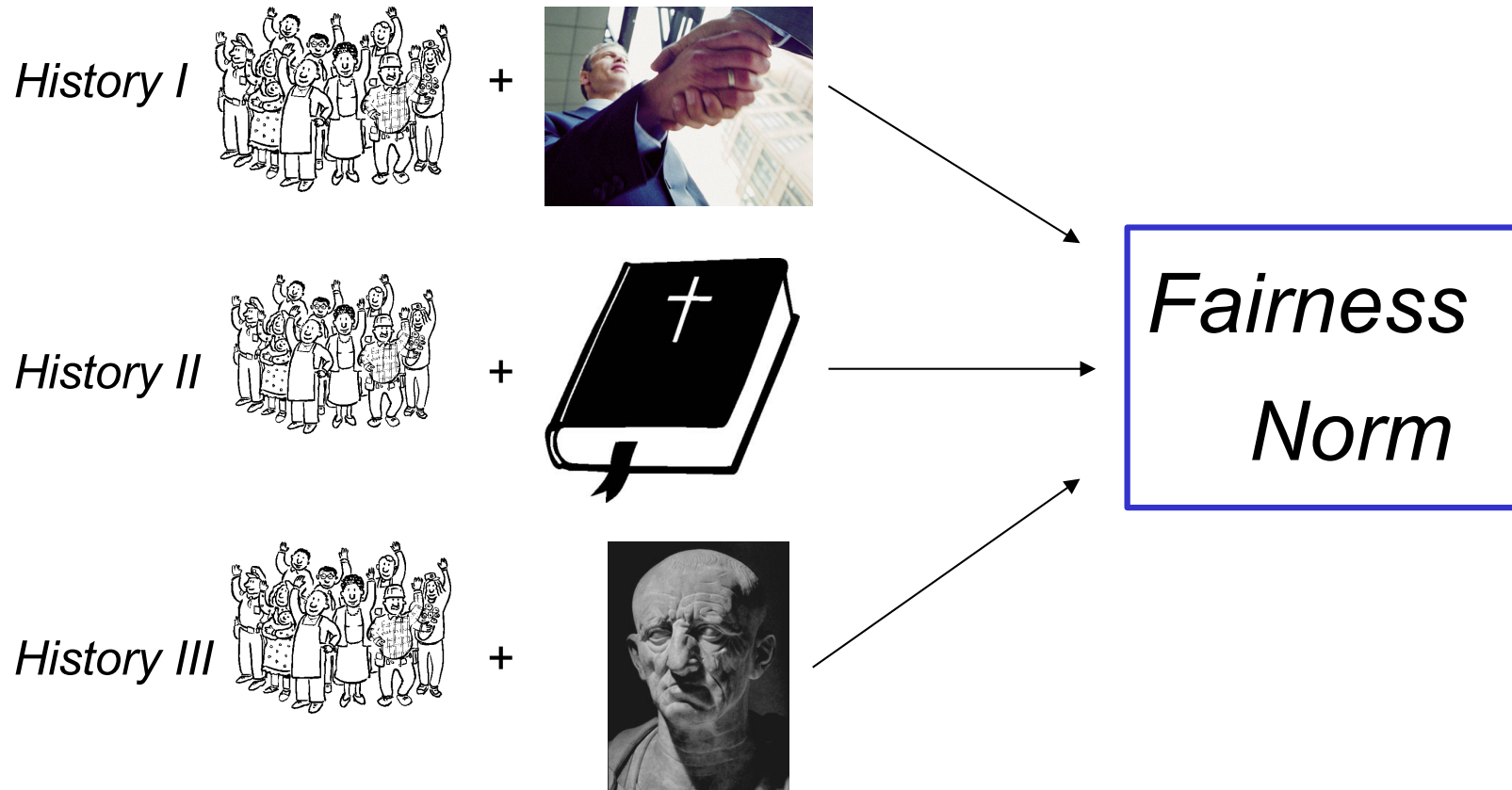
Offer & Acceptance dependent on:

- Culture
- Did player 1 earn the reward?
- Was she responsible for the offer?

Explanation: social norms

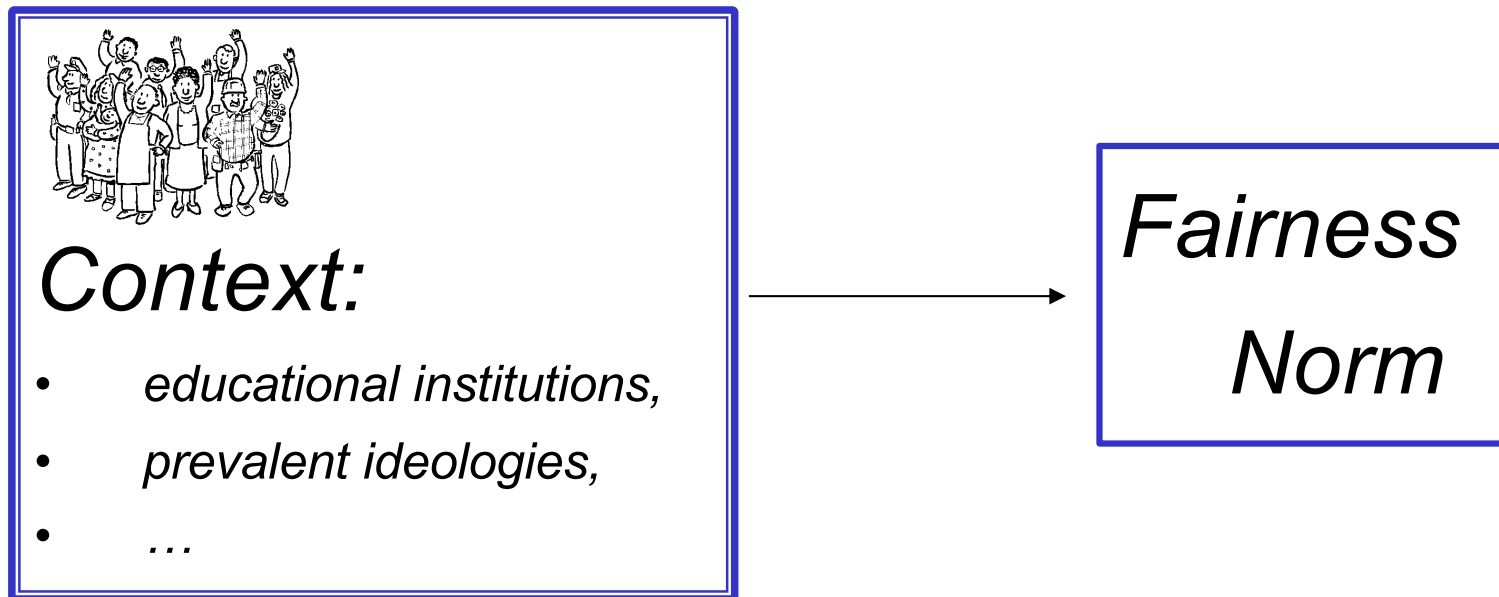
When Individualists Accounts are Incomplete

- Multiple realisation



When Individualists Accounts are Incomplete

- Multiple realisation
- Context-dependence



When Individualists Accounts are Incomplete

Ontologically correct:

- Every particular social norm constituted by particular interaction of individuals in particular circumstances
- TOKEN-TOKEN relationship!

Methodologically relevant:

- Social scientists aim at describing *in general* the relation between social norms and human behaviour
- TOKEN-TOKEN relationship only gives us specific historical narratives
- do not want to trade generality for methodological indiv.

Learning Results

- Name some social science methods and describe their special characteristics
- Distinguish qualitative and quantitative data and apply strategies to improve the production of the former
- Describe the concept of a case study and apply strategies for selecting cases
- Describe the concept of interpretation and discuss the relevance of interpretative methods for a distinct social science methodology
- Distinguish naturalistic vs. non-naturalistic methodology of the social sciences
- Apply strategies for improving interpretative methods
- Distinguish methodological individualism vs. holism and discuss arguments for the latter